Yacoub Derradji

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Summary

Third-year Mechatronic Systems Engineering student (dual in AI Systems Engineering) who designs robotic mechanisms and actuators and is interested in humanoid robotics. Seeking a 4-month co-op for Summer 2027.

Education

Western University Sept. 2024 – Apr. 2029 (Expected)
BESc, Mechatronic Systems Engineering, Dual in AI Systems Engineering *London, ON*
- Cumulative GPA: 3.8 (Dean's Honour List)
- Coursework: Manufacturing Processes, Mechanics of Materials, Dynamics, Computational Methods, Algorithms & Data Structures, Introduction to Machine Learning (in progress, Fall 2026)

Experience

Undergraduate Researcher May 2026 – Present
Advanced Robotics & Mechatronics (ARM) Lab, Western University *London, ON*
- Redesigned 7 parts of a miniaturized planetary-gear magnetorheological (MR) clutch, the per-joint actuator of a multi-finger robotic hand, within a 20 mm OD, 10 mm wide envelope
- Traced a torsional shaft fracture and an unmanufacturable keyway to one stress-concentration root cause; fixed both by upsizing the shaft from 1 to 2 mm and implementing a D-profile shaft/bore coupling
- Produced the 7-drawing SolidWorks manufacturing set under ISO 2768-m; in design review, caught a bearing-seat fit error and re-specified the seat to +0.02/0 mm
- Sourced all 7 parts across five manufacturing platforms, vetting vendors on gear-cutting capability and material compatibility; quotes pending approval for order

Mechanical Lead, BotBrawl Combat Robot Team Aug. 2026 – Present
WE First, Western University *London, ON*
- Lead mechanical design and build of the club's combat robot for the 2026–27 BotBrawl season, organizing the mechanical sub-team

Planning and Controls Developer Oct. 2025 – Dec. 2025
Western Engineering Autopilot (WEAP) *London, ON*
- Made full-system ROS 2/Gazebo simulation startup 6x faster (3 min to 30 s) by restructuring the launch architecture for the planning and control subsystems

Electrical Team Member Sept. 2024 – Mar. 2025
Western Formula Racing (FSAE) *London, ON*
- Created the tractive system schematic and connector pinouts for the team's electric race car

Projects

Humanoid Neck-Tilt Mechanism (Baker Bot) | *Onshape, Prototyping* May 2026 – Aug. 2026
- Developed a 2-DOF neck-tilt mechanism for Baker Bot, a torso-up humanoid (WE First), taking it from concept to working hardware: two angled 40 kg servos drive ball-jointed pushrods against a rear fixed link for pitch and limited roll
- Did the full Onshape design (off-the-shelf rod ends, custom 3D-printed mounts), prototyped and iterated it, implemented basic nodding, and integrated it with a teammate's rotation base before handing off to the software and torso teams

Off-Road RC Vehicle | *SolidWorks, Mechanism Design, Embedded Control* Dec. 2025 – Apr. 2026
- Designed and built a full vehicle from custom 3D-printed parts (chassis, modular FWD/4WD drivetrain, steering, suspension) under a $200 budget, with no kit; integrated Arduino control with motor drivers and servo steering
- First test drives exposed suspension and steering problems that kept it from driving; revised suspension arm lengths and steering knuckles to get it running

Tendon-Driven Robotic Gripper | *Prototyping, Grasping* Sept. 2025 – Present
- Built a 3-finger gripper prototype by adapting an open-source finger design for rubber-band and string tendons, with a 3-servo palm

Skills & Certifications

CAD & DFM: SolidWorks, Onshape, GD&T, ISO 2768 Tolerancing, Fits & Clearances, DFM, Engineering Drawings, Vendor Quoting, 3D Printing
Robotics & Mechanisms: Linkage Design, Gear Trains, Shaft/Coupling & Bearing Design, Servo Control, ROS 2, Gazebo, MATLAB, Simulink
Electrical & Embedded: Arduino, ESP32, Motor Drivers, PWM Control, Schematics & Pinouts, Soldering
Programming: Python, C/C++, Git/GitHub, Java

Certifications & Awards: Certified SolidWorks Associate (CSWA); 3rd Place, Western Engineering Competition, Junior Division (Nov. 2025); Top 3 of 72 Teams, Ivey Case Competition (Mar. 2025)